

Question	Marking Guidance	Mark	Comments
10.1	1. To break actinomyosin (bridges); 2. To move/bend the myosin head/arm; 3. (So) actin (filaments) are moved (inwards); 4. For active transport of calcium <u>ions</u> (into the sarcoplasmic/endoplasmic reticulum);	2 max (2 x AO1)	1. Accept 'to form actinomyosin'. 1. Accept 'to detach or attach myosin and actin'. 1. Reject reference to 'active site'. 2. Accept 'to change shape of myosin head/arm'. 2. Accept ADP/Pi moves (myosin) head/arm. 2. Accept powerstroke/pivot /recocks etc. for movement. 3. Ignore 'sliding'.
Question	Marking Guidance	Mark	Comments
10.2	1.35 OR 1.4;	1 (AO2)	Accept numbers after 1.35.

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10.3	- Fast (skeletal muscle) fibres used during short-term/intense exercise; - Slow (skeletal muscle) fibres used during long(er)-term exercise; - Creatine used to form phosphocreatine; (Phosphocreatine) combines with ADP to form ATP; (Carbohydrate/glucose) stored as glycogen OR - Glycogenesis; - Glycogen hydrolysed to glucose OR - Glycogenolysis; - Glucose for respiration;	5 max (5 x AO2)	Max 3 marks from mark points 3, 4, 5, 6 and 7. 1 and 2. Accept examples of short-term/intensity exercise e.g. sprint and longer-term/endurance exercise, e.g. marathon or low(er) intensity exercise. 1. Accept fast twitch fibres for fast (skeletal muscle) fibres 2. Accept slow twitch fibres for slow (skeletal muscle) fibres. 7. Accept glycolysis for respiration.

Question	Marking Guidance	Mark	Comments
10.4	1. (More) glucose enters (muscle) cells; 2. (Glucose/fatty acids enter by) facilitated diffusion; 3. Fatty acids used in Krebs cycle; 4. Respiration provides (more) ATP;	3 max (3 x AO2)	Max 2 marks if only in context of glucose or only in context of fatty acids. 1. Ignore more glucose leaves (liver) cells. 1. Reject glucose enters mitochondria. 2. Accept active transport or cotransport. 3. Ignore gluconeogenesis. 4. Accept in context of glucose or fatty acids. 4. Accept for fatty acids 'Krebs cycle produces ATP'.

Question	Marking Guidance	Mark	Comments
10.5	1. Increase in CO ₂ detected by chemoreceptors; 2. Send (more) impulses to cardiac centre OR Send (more) impulses to the medulla; 3. More impulses (from centre/medulla) along sympathetic pathway/neurones/nerves OR Fewer impulses (from centre/medulla) along parasympathetic/vagus pathway/neurones/nerves; 4. (To) SAN;	4 (1 x AO1, 3 x AO2)	1. Accept increase in acidity/H ⁺ or decrease in pH for increase in CO ₂ . 1. Ignore location of chemoreceptors. 2 and 3. Reject reference to 'an/one impulse' once only. 2 and 3. Reject 'signals', 'messages' for 'impulses' once only. 2 and 3. Accept 'action potentials' for impulses.